

INSTRUCTIONS TO THE CANDIDATES

- ❖ Do not use sketch pens/ high –lighter pens/ color pencils etc., in underlining / highlighting certain points in the answer or while drawing sketches etc.,
 - ❖ Check the Answer booklet thoroughly. The defective answer booklet may be returned to the invigilator before starting the examination.
 - ❖ Ensure that the OMR Barcode sheet is properly secured to the answer booklet.
 - ❖ Do not write on or tamper the Barcode. If you do so, your booklet will not be evaluated.
 - ❖ The candidate is prohibited from writing his/ her NAME / ADDRESS/ PHONE NUMBER / EMAIL ID / Religion or any other matter on any part of the answer booklet / drawing sheet/ graph sheet/ addressing the examiner in any manner whatsoever in the answer booklet. The candidate is not supposed to tear a page in full or part of the booklet.
- Non-adherence with this instruction shall result in candidate being booked under malpractice.**
- ❖ Answer should be written on both sides of the paper. Candidate should write in all 25 lines in each page. It is not necessary to begin each answer in a fresh page.
 - ❖ Candidate should write only question number in the margin.
 - ❖ Candidate should return the answer booklet to the invigilator before leaving the examination hall.
 - ❖ **ADDITIONAL SUPPLEMENT OF ANSWER SHEETS WILL NOT BE ISSUED.**

INSTRUCTIONS TO THE EXAMINERS

- ❖ Invigilators should ensure that the above instructions are strictly followed by the candidates.
- ❖ After verifying the particulars on the OMR sheet, invigilator should sign in the box specified.
- ❖ Do not crumple or cut the sides of the sheets.
- ❖ Do not write, mark or damage the timing tracks or Barcode.

5 A

CFG into CNF

Context free grammar $\rightarrow$ Chomsky Normal Form.

$$\begin{array}{l} A \rightarrow a \\ B \rightarrow AB \end{array}$$

CFG

$$S \rightarrow \overset{\checkmark}{A}\overset{\checkmark}{B} / \overset{\checkmark}{a}$$

$$A \rightarrow \overset{\checkmark}{B}\overset{\checkmark}{C} / \overset{\checkmark}{b} / \overset{\checkmark}{\epsilon}$$

$$B \rightarrow aB / C$$

$$C \rightarrow aC / B / \epsilon$$

1) generate nulls.

$$S \rightarrow AB / a$$

$$A \rightarrow BC / b / \epsilon$$

$$B \rightarrow aB / C$$

$$C \rightarrow aC / B.$$

Remove useless production,

$$S \rightarrow AB/a$$

$$A \rightarrow BC/b$$

$$B \rightarrow aB/c \quad \begin{matrix} E \\ \Rightarrow aCB \\ D \end{matrix}$$

$$C \rightarrow aC/B \quad \begin{matrix} D \\ \Rightarrow aBC \\ E \end{matrix}$$

$$D = aB$$

$$E = aC$$

$\Rightarrow$

$$S \rightarrow AB/a$$

$$A \rightarrow BC/b$$

$$B \rightarrow DD/EB$$

$$C \rightarrow EE/DC$$

$$S \rightarrow AB/a$$

$$A \rightarrow BC/b$$

$$B \rightarrow DD/EB$$

$$C \rightarrow EE/DC$$

$$D \rightarrow aB$$

$$E \rightarrow aC$$

S B

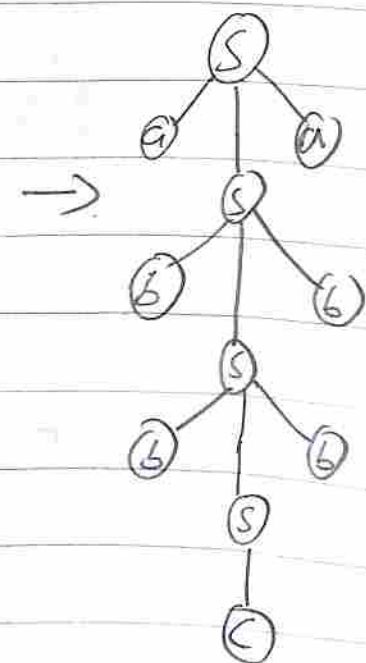
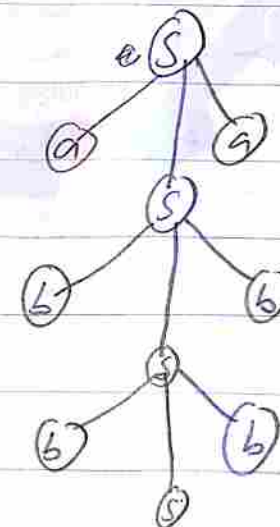
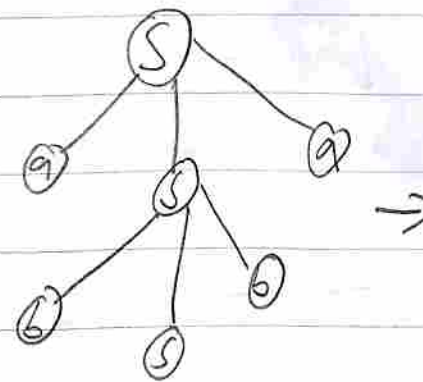
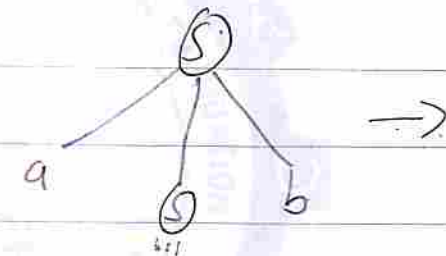
$$S \rightarrow aSa / bSb / c$$

$$w = abbcbba$$

$$\Rightarrow S \rightarrow aSa$$

$$S \rightarrow bSb$$

$$S \rightarrow c$$

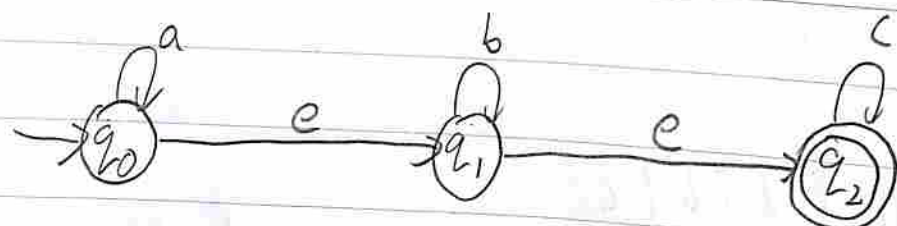


abbcbba

4 A

4 A

ϵ -NFA



ϵ NFA TT

$\epsilon(q_0) \rightarrow q_0, q_1, q_2$

$\epsilon(q_1) \rightarrow q_1, q_2$

$\epsilon(q_2) \rightarrow q_2$

ϵ NFA

		a	b	c
q_0	q_0	q_0	-	-
q_1	q_1	-	q_1	-
q_2	q_2	-	-	q_2

	ϵ	a	ϵ
$\rightarrow q_0$	q_0, q_1, q_2	q_0	q_0, q_1, q_2
q_1	q_1, q_2	-	-
$* q_2$	q_2	-	-

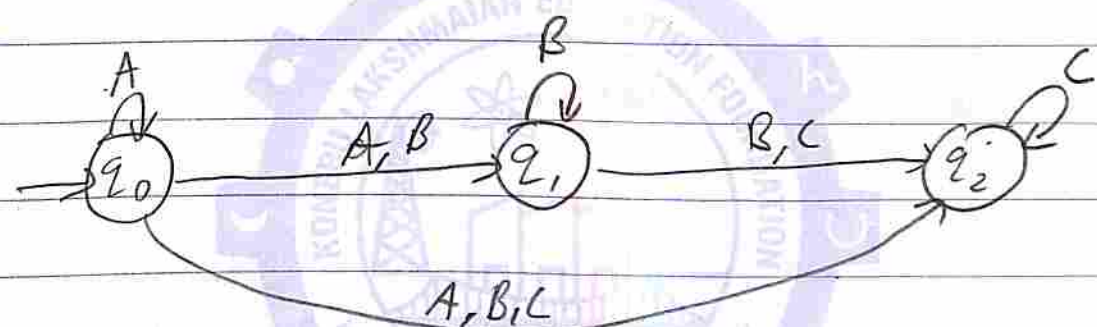
	ϵ	b	ϵ
$\rightarrow q_0$	q_0, q_1, q_2	q_1	q_1, q_2
q_1	q_1, q_2	q_1	q_1, q_2
$* q_2$	q_2	-	-

	ϵ	c	ϵ
q_0	-	-	-
q_1	-	-	-
q_2	-	-	-

	ϵ	c	ϵ
$\rightarrow q_0$	q_0, q_1, q_2	q_2	q_2
q_1	q_1, q_2	q_2	q_2
$* q_2$	q_2	q_2	q_2

NFA TT

	A	B	C
$\rightarrow q_0$	q_0, q_1, q_2	q_1, q_2	q_2
q_1	-	q_1, q_2	q_2
$* q_2$	-	-	q_2



4 A

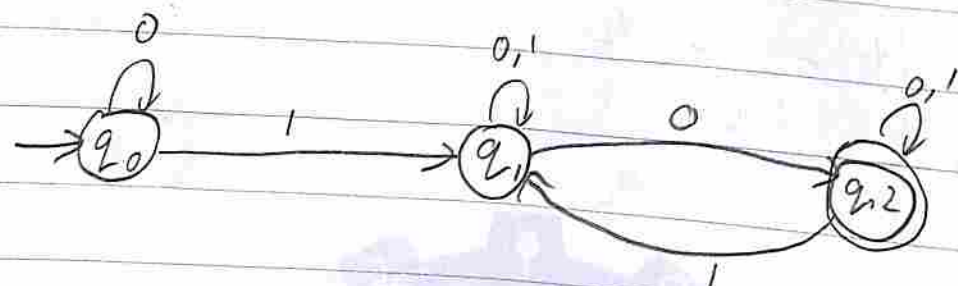
4 B

$L = \{a^p \mid p \text{ is prime}\}$ regular or not.

we use the pumping Lemma to find out.

2 A

NFA $\rightarrow$ PFA



Transition Table.

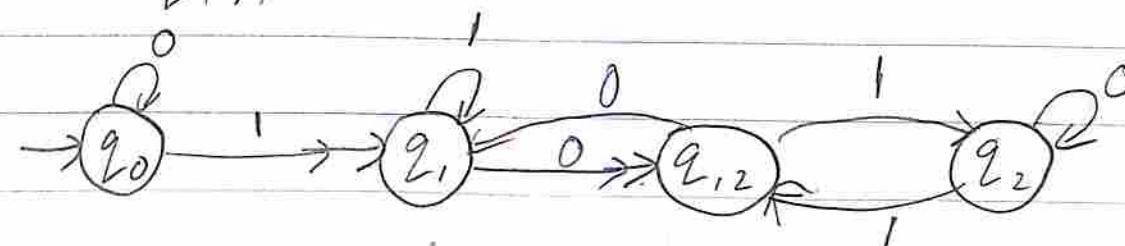
	0	1
q_0	q_0	q_1
q_1	q_1, q_2	q_1, q_2
q_2	q_2	q_1, q_2

Make $q_{1,2}$

	0	1
q_0	q_0	q_1
q_1	$q_{1,2}$	$q_{1,2}$
q_2	$q_{1,2}$	$q_{1,2}$
$q_{1,2}$	$q_{1,2}$	$q_{1,2}$

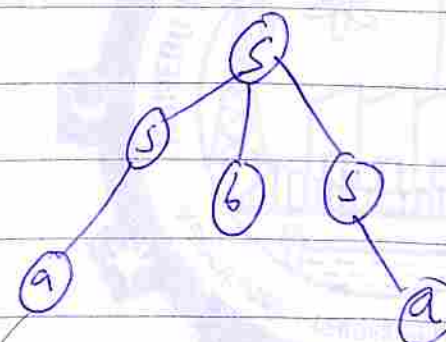
$\leftarrow$ DFA

DFA



2 B

$S \rightarrow SbS/a$



$\Rightarrow a b a$

2 B

RE $\rightarrow r = (ab+9)^* (aa+b)$

$aa+9 \rightarrow \{\epsilon, aa, b, aab, ba9\}$

$ab+9 \rightarrow \{\epsilon, a9, a, aba, a9b\}$

2 C

$S \rightarrow AB/CA$

$B \rightarrow BC/AB$

$A \rightarrow a$

$C \rightarrow aB/b$

1 A

NFA

NON Deterministic Finite Automata.

Every state in the FA may not always need every transition for each of elements in language.

1 B

Regular Expression

- Concatenation
- Union
- Multiplication

please
pls

1 C

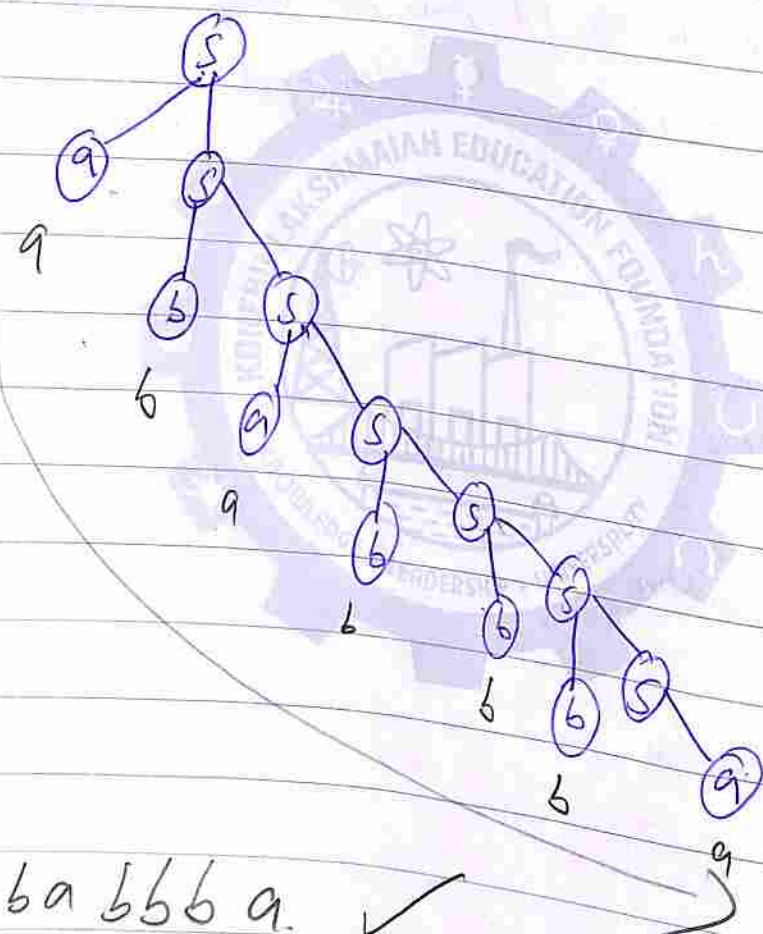
Pumping lemma

It is used to check if the given expression is a regular expression or not.

1 D

$S \rightarrow aS / bS / a / b / \epsilon$

$w = ababbb a.$



1 E

Grammar,

It is the use of extras to make the language is understandable and more better

1 E

CFG

Context Free Grammar.

- Terminals
- variables.



